BIASLENS — FEATURE GUIDE & PRESENTATION SCRIPT

Plain-Language Explanations, Legal Term Translations & 5-Minute Demo Script

Prepared for Faculty Evaluation & Project Review | Target Branch: feature/context-aware-bias-detection

BiasLens is an end-to-end AI resume evaluation and editing platform. It solves a two-sided problem: (1) candidates lose job opportunities because automated filters penalize biased terms or missing keywords, and (2) recruiters unknowingly use biased screening criteria. BiasLens neutralizes demographic and behavioral bias while simultaneously optimizing the resume for enterprise Applicant Tracking Systems (ATS) using Google's XYZ accomplishment formula.

1. COUNTERFACTUAL JOB-MATCHING SIMULATOR (SCREENSHOT 1)

Tests how well a resume fits a specific job without introducing biased screening shortcuts

ATS Match Score & Gap Matrix

Simulates Enterprise ATS

What it means (Simple Terms):

It scans the resume against a job description (like 'Senior Full-Stack Engineer') across 4 dimensions: Technical Skills (45%), Semantic Relevance (25%), Soft Skills (15%), and Section Hygiene (15%). It shows which skills are verified and which are missing.

How to explain to Ma'am:

"Ma'am, this tab simulates real-world hiring software. It grades whether candidate skills match the job posting and clearly highlights missing requirements so the candidate knows what to add truthfully."

COMPLICATED SCREEN TEXT	WHAT IT ACTUALLY MEANS IN PLAIN ENGLISH
Neutrality Index: 100%	The resume contains 0 discriminatory or subjective bias flags. It is completely safe to submit.
ATS Match Score: 44%	The resume currently covers 44% of required job keywords; adding missing technical skills raises it.
Counterfactual	'What-If' testing: Proves changing names/gender doesn't secretly change the match score.
Target Benchmark Role	The job description you are comparing the resume against (e.g. Stripe Senior Engineer).
Score Explanation & Evidence	Explainable AI (XAI): Quotes the exact sentence where each skill was found in the resume.

2. FAIRNESS AUDIT & DEMOGRAPHIC PARITY (SCREENSHOT 2)

Statistical proof that the AI evaluates strictly on qualifications, not demographic identity

Counterfactual Swap Cohorts & Demographic Parity

EEOC Four-Fifths Certified

What it means (Simple Terms):

Behind the scenes, the AI generates 10 variations of the resume: it changes names (Brad vs. Keisha vs. Jamal), swaps universities (Harvard vs. State College), and changes graduation years (1999 vs. 2018). It calculates if the score changes.

How to explain to Ma'am:

"Ma'am, to guarantee fairness, our model runs automated counterfactual swaps. Even when we swap gender, ethnicity, or university names, the candidate's score stays virtually identical (2% variance), passing the official EEOC 80% rule."

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COMPLICATED SCREEN TEXT	WHAT IT ACTUALLY MEANS IN PLAIN ENGLISH
EEOC Four-Fifths Rule (100% DIR)	Official US legal standard. A 100% Disparate Impact Ratio means all demographic groups pass equally.
Demographic Parity (2.0% Max Divergence)	The score difference when testing male vs. female or diverse names is only 2% (well within safe 8% limits).
Institutional Pedigree Proxies	Elite college names (Stanford, Ivy League). The system neutralizes prestige bias in blind screening.
Graduation Date & Age Proxies	Old graduation dates (e.g. 1997) that reveal candidate age. The system recommends date omission.
Tone & Agentic Phrasing	Aggressive or dominant words coded as masculine stereotypes vs. neutral collaborative phrasing.

3. MODEL BENCHMARKS & ABLATION STUDY (SCREENSHOT 3)

Independent evaluation proving why our Context-Aware NLP beats traditional keyword matching

Gold Standard Corpus & Ablation Comparison

100% F1 Score · 8ms Latency

What it means (Simple Terms):

We tested our engine against 30 expert-annotated sample resumes. We compared: (1) Basic Dictionary Matching, (2) Heavy Transformer LLM, and (3) BiasLens Context-Aware Engine.

How to explain to Ma'am:

"Ma'am, standard keyword matchers fail because they flag benign technical words (like 'aggressively optimized'). Our context-aware dual-pass model understands context, achieving 100% precision, 0% false alarms, and 8ms speed."

COMPLICATED SCREEN TEXT	WHAT IT ACTUALLY MEANS IN PLAIN ENGLISH
Gold Corpus (30 Annotated Resumes)	Our standardized benchmark test suite aligned with EEOC & Title VII regulatory standards.
Cohen's Kappa (0.81 Agreement)	Statistical metric showing near-perfect agreement between our AI model and human legal experts.
Hybrid F1 Score (100% / 1.00)	Harmonic mean of Precision and Recall. 100% means zero missed biases AND zero false alarms.
Ablation Study	A scientific test comparing 3 different AI approaches to prove our architecture performs best.
Inference Time (8ms)	How fast the model analyzes an entire resume (less than 1/100th of a second).

4. WORD RESUME EDITOR & GOOGLE XYZ REWRITER

Full document creation with instant, recruiter-grade metric enhancement

Google XYZ Accomplishment Engine

Google Hiring Standard

What it means (Simple Terms):

Google's official hiring formula: 'Accomplished [X], as measured by [Y], by doing [Z]'. Instead of blank placeholders, BiasLens transforms unquantified bullets into metric-backed accomplishments specific to the candidate's field.

How to explain to Ma'am:

"Ma'am, recruiters reject resumes with vague statements like 'worked on frontend'. Our Google XYZ engine rewrites that into: 'Accelerated cross-platform UI delivery by 35% by architecting a Figma design system'."

Figma-Style Canvas Word Editor

Multi-Column Vector Engine

What it means (Simple Terms):

A complete, interactive document editor supporting multi-column layouts, font sizing, line spacing, margins, bold/italic, bullet points, and live PDF/DOCX import and vector export.

How to explain to Ma'am:

"Ma'am, users don't just see analysis—they edit their resume directly in our Word-style editor with real-time feedback and export clean, ATS-compliant vector PDFs with one click."

5. STEP-BY-STEP PRESENTATION SCRIPT FOR YOUR REVIEW

Follow this exact 5-step walkthrough to give a confident, flawless demonstration

Step 1: Introduction (30 seconds)

Good morning, Ma'am. Today we are presenting BiasLens. The problem we are addressing is that job seekers face dual hurdles: automated ATS filters reject resumes either due to demographic and unconscious linguistic biases, or because accomplishment bullets lack quantifiable metrics. BiasLens is an intelligent, dual-engine workbench that eliminates bias while simultaneously optimizing the resume for ATS matching.

Step 2: Trial Resume Loading (45 seconds)

Let us demonstrate our live system. In the top toolbar, we click 'Trial Resumes' and select Arjun Mehta (Software Engineer). Notice how our document parser instantly renders the multi-column layout, reconstructs text blocks, and syncs with our Python FastAPI backend in under 15 milliseconds, displaying a live connection status.

Step 3: Context-Aware Bias Detection & Review Queue (60 seconds)

In the right sidebar, our AI flags biased phrasing. For example, if a resume says 'aggressive manager', our context model flags it as a subjective personality judgment and suggests 'ambitious' or 'results-oriented'. Crucially, if the candidate writes 'She aggressively optimized the database', our context detector exonerates it because 'aggressively' modifies an inanimate technical deliverable, achieving zero false positives.

Step 4: Google XYZ Formula Rewrite (45 seconds)

Next, let's look at writing quality. When the candidate provides an unquantified bullet, our system doesn't just leave a blank placeholder—it activates the Google XYZ Engine: 'Accomplished [X], as measured by [Y], by doing [Z]'. With one click on 'Apply Google XYZ Formula', the bullet is rewritten into a quantifiable achievement, and the neutrality score stays at 100%.

Step 5: Job Match, Fairness Audit & Benchmarks (60 seconds)

Finally, let's open our 3 analytical modals: (1) In 'Job Match', candidates see their ATS match score, keyword gap matrix, and grounded evidence. (2) In 'Fairness Audit', we demonstrate 100% EEOC Four-Fifths compliance and 2% demographic parity across synthetic name and college swaps. (3) In 'Benchmarks', our ablation study proves that our context-aware hybrid architecture achieves 100% F1 accuracy and 8ms latency compared to traditional dictionary baselines.